

Secretary Problem Research

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February 7, 2022

Contents

1 Abstract Error SP	1
1.1 Basic Definitions	1
1.2 Algorithmic Representations	2
1.3 Calculation under a relaxed condition	3
1.4 relaxed model PI	5
1.4.1 calculation 1 (in meeting)	5
1.4.2 Calculation 2	6
2 Relaxed SP Prophet Inequality	6
2.1 Lower Bound of $P^{\max}$	6
2.2 Upper Bound of $P^{\max}$	7
2.3 Responses	8
2.3.1 Response to (2.1)	8
2.3.2 Response to (2.2)	8
3 Calculation with error involved	9

Lecture 1: abstract error SP

Sun 30 Jan 2022 15:42

1 Abstract Error SP

1.1 Basic Definitions

$i \approx j$ means that i and j are in-distinguishable in the first observation.

- (reflexivity) $i \approx i$
- (symmetry) $i \approx j \Leftrightarrow j \approx i$
- (connectedness) $i \approx j, r_i < r_j \Rightarrow \forall k, l [r_k, r_l \in [r_i, r_j] \Rightarrow r_k \approx r_l]$

1.2 Algorithmic Representations

Algorithm 1 Oct 14 2021

```
1: for  $i \in [n]$  do
2:    $\alpha \leftarrow$  the current item to be decided
3:   if  $i < r$  then  $\triangleright$  Phase 1
4:     if  $\exists \beta \in O_i. \alpha <_i \beta$  then  $\triangleright \alpha$  is not a maximum
5:       Decline this item
6:     else if  $\alpha \in E_i$  then
7:       Decline this item
8:     else
9:       Examine this item
10:    end if
11:  else if  $i < n$  then  $\triangleright$  Phase 2
12:    if  $\exists \beta \in O_i. \alpha <_i \beta$  then
13:      Decline this item
14:    else
15:      if  $\forall \beta \in O_i. \beta \leq_i \alpha$  then  $\triangleright \alpha$  is the absolute maximum
16:        Accept this item
17:      else if  $\alpha \in E_i$  then
18:        Decline this item
19:      else
20:        Examine this item
21:      end if
22:    end if
23:  else
24:    Accept this item
25:  end if
26: end for
```

Algorithm 2 Game Process, Relaxed

```

1:  $[n], r_i$  and  $\approx$  relation are given according to definition
2:  $O_1 = \{1\}$ 
3:  $E_1 = \emptyset$ 
4:  $\alpha_1 = 1$ 
5:  $\Theta_1 = \emptyset$ 
6: for  $i \in [n]$  do
7:    $d_i \leftarrow$  player's decision
8:   if  $d_i = \text{Accept}$  then
9:     Game terminates with reward  $v(\alpha_i)$ 
10:  else if  $d_i = \text{Decline}$  then
11:     $O_{i+1} \leftarrow O_i \cup \{i+1\}$ 
12:     $E_{i+1} \leftarrow E_i$ 
13:     $\alpha_{i+1} \leftarrow i+1$ 
14:  else if  $d_i = \text{Examine}$  then
15:     $O_{i+1} \leftarrow O_i$ 
16:     $E_{i+1} \leftarrow E_i \cup \{i\}$ 
17:     $\alpha_{i+1} \leftarrow \alpha$ 
18:  end if
19:   $\Theta_{i+1} \leftarrow \{(\beta, \gamma) \in O_{i+1}^2 : r_\beta \leq r_\gamma, \beta \not\approx \gamma\} \cup \{(\beta, \gamma) \in E_{i+1}^2 : r_\beta \leq r_\gamma\}$ 
20: end for

```

1.3 Calculation under a relaxed condition

We give the expected value of ALG on the relaxed model (only one examine is required to obtain precise comparisons). Let α denote the selected item.

$$E[\text{OPT}] = m \quad (1)$$

$$E[\text{ALG}] = \sum_{t=1}^m t P[v(\alpha) = t] \quad (2)$$

$$P[v(\alpha) = t] = \sum_{i=r}^n P[\alpha = i \mid v(i) = t] P[v(i) = t] \quad (3)$$

$$+ P[\text{reach } n \cap \text{not accept } n] P[v(n) = t] \quad (4)$$

$$+ P[\text{reach } n-1 \cap \text{examine and decline } n-1] P[v(n-1) = t] \quad (5)$$

The two added terms are cases where the items are exhausted. If n is reached then it must be selected. If $n-1$ is examined then it must be selected also.

$$P[v(i) = t] = 1/m, \forall i \forall t \quad (6)$$

$$P[\alpha = i \mid v(i) = t] = P[\text{reach } i \mid \max_{j < r} v(j) \leq t] P[\max_{j < r} v(j) \leq t] \quad (7)$$

$$= \sum_{s=1}^t P[\text{reach } i \mid \max_{j < r} v(j) = s] P[\max_{j < r} v(j) = s] \quad (8)$$

$$P[\text{reach } n \cap \text{not accept } n] = \sum_{s=t+1}^m P[\text{reach } n \mid \max_{j < r} v(j) = s] P[\max_{j < r} v(j) = s] \quad (9)$$

$$P[\text{reach } n-1 \cap \text{examine and decline } n-1] = \sum_{s=t+1}^m \mathbf{1}_{s \leq t+\delta} P[\text{reach } n-1 \mid \max_{j < r} v(j) = s] P[\max_{j < r} v(j) = s] \quad (10)$$

$$P[\max_{j < r} v(j) = s] = P[\max_{j < r} v(j) \leq s] - P[\max_{j < r} v(j) \leq s-1] \quad (11)$$

$$= \left(\frac{s}{m}\right)^{r-1} - \left(\frac{s-1}{m}\right)^{r-1} \quad (12)$$

$$P[\text{reach } i \mid \max_{j < r} v(j) = s] = \sum_{(a_1, \dots, a_k): a_j \in \{1, 2\}, \sum a_j = i-r} \prod_{j=1}^k \mathbf{1}_{\delta < t} \left(\mathbf{1}_{a_j=1} \frac{s-\delta-1}{m} + \mathbf{1}_{a_j=2} \frac{\delta}{m} \right) + \mathbf{1}_{\delta \geq t} \frac{t-1}{m} \quad (13)$$

$$= \sum_{x=0}^{\lfloor \frac{i-r}{2} \rfloor} \left(\mathbf{1}_{\delta < t} \left(\frac{\delta}{m} \right)^x \left(\frac{s-\delta-1}{m} \right)^{i-r-2x} + \mathbf{1}_{\delta \geq t} \left(\frac{t-1}{m} \right)^x 0^{i-r-2x} \right) \frac{(i-r-x)!}{x!(i-r-2x)!} \quad (14)$$

Explanation: There are $i - r$ steps to go from r to i . Each decline will go 1 step, and each examine+decline will go 2 steps. We calculate the probability to take each possible path and then sum them up. In the second equation, x is the number of items examined going from r to i .

Lecture 2: Relaxed SP

Sun 30 Jan 2022 15:50

1.4 relaxed model PI

1.4.1 calculation 1 (in meeting)

$$\begin{aligned}
& \sum_{i=1}^n \mathbb{P}[\text{accept } i \cap i \text{ best}] \\
&= \sum_{i=1}^n \mathbb{P}[\text{accept } i \mid i \text{ best}] \mathbb{P}[i \text{ best}] \\
&= \frac{1}{n} \sum_{i=1}^n \mathbb{P}[X_1, \dots, X_{i-1} < T \cap X_i \geq T \mid i \text{ best}] \\
&= \frac{1}{n} \sum_{i=1}^n \mathbb{P}[X_1, \dots, X_{i-1} < T \mid X_i \geq T \cap i \text{ best}] \mathbb{P}[X_i \geq T \mid i \text{ best}]
\end{aligned}$$

Note that $\mathbb{P}[X_i \geq T \mid i \text{ best}] = 1 - F_{X_{\max}}(T) = 1 - T^n$. Also,

$$\begin{aligned}
& \mathbb{P}[X_1, \dots, X_{i-1} < T \mid X_i \geq T \cap i \text{ best}] \\
&= \int_T^1 \mathbb{P}[X_1, \dots, X_{i-1} < T \mid i \text{ best} \cap X_i = x] f_X(x \mid X_i \geq T \cap i \text{ best}) dx \\
&= \int_T^1 \mathbb{P}[X_1, \dots, X_{i-1} < T \mid X_1, \dots, X_{i-1} < x] f_{X_{\max}}(x \mid X_{\max} \geq T) dx \\
&= \int_T^1 \left(\frac{T}{x}\right)^{i-1} \frac{nx^{n-1}}{1 - T^n} dx \\
&= \frac{nT^{i-1}}{1 - T^n} \int_T^1 x^{n-i} dx \\
&= \frac{nT^{i-1}}{1 - T^n} \frac{1 - T^{n-i+1}}{n - i + 1}
\end{aligned}$$

Combining all terms,

$$\begin{aligned}
& \sum_{i=1}^n \mathbb{P}[\text{accept } i \cap i \text{ best}] \\
&= \frac{1}{n} \sum_{i=1}^n \frac{nT^{i-1}}{1 - T^n} \frac{x^{n-i+1}}{n - i + 1} (1 - T^n) \\
&= \sum_{i=1}^n T^{i-1} \frac{1 - T^{n-i+1}}{n - i + 1}
\end{aligned}$$

1.4.2 Calculation 2

We have n items with weights $X_1, \dots, X_n \sim \text{Unif}(0, 1)$. Observed values are $\tilde{X}_i \sim \text{Unif}(X_i - \delta, X_i + \delta)$. We adopt a thresholding scheme for ALG: We examine an item if $\tilde{x}_i > T$ and accept if $x_i > T$.

If the player reaches X_n , then X_n must be accepted.

$$\begin{aligned}
P[X_{\max} \text{ selected}] &= \sum_{i=1}^{n-1} P[X_1, \dots, X_{i-1} < T \cap X_i \geq T \cap X_i \text{ is max}] \\
&\quad + P[X_1, \dots, X_{n-1} < T \cap X_n \text{ max}] \\
&= \sum_{i=1}^{n-1} \int_0^1 P[X_1, \dots, X_{i-1} < T \cap X_i \geq T \cap X_{i+1}, \dots, X_n \leq X_i \mid X_i = x] f_{\text{Unif}(0,1)}(x) dx \\
&\quad + \frac{1}{n} \int_0^1 P[X_n < T \mid X_n = x] f_{\text{Unif}(0,1)}(x) dx \\
&= \sum_{i=1}^{n-1} \int_T^1 P[X_1, \dots, X_{i-1} < T \cap X_{i+1}, \dots, X_n \leq x] f_{\text{Unif}(0,1)}(x) dx \\
&\quad + \dots \\
&= \sum_{i=1}^{n-1} \int_T^1 T^{i-1} x^{n-i} dx + \int_0^T x^{n-1} dx + \int_T^1 x^{n-1} \frac{T}{x} dx \\
&= \sum_{i=1}^{n-1} \frac{1}{n-i+1} T^{i-1} (1 - T^{n-i+1}) + \frac{1}{n} T^n + T \frac{1 - T^{n-1}}{n-1}
\end{aligned}$$

2 Relaxed SP Prophet Inequality

2.1 Lower Bound of $P^{\max}$

In this section, we provide a lower bound of $P^{\max}$, where

$$P^{\max} = \max_{T \in [0,1]} \sum_{i=1}^n \frac{1}{n-i+1} T^{i-1} (1 - T^{n-i+1}).$$

The optimal T^* can be solved by the optimization problem. Let's show the lower bound of $P^{\max}(T^*)$. Let $T = \frac{n}{n+1}$, then we have

$$\begin{aligned}
P^{\max}(T^*) &\geq P^{\max}(T) \\
&= \sum_{i=1}^n \frac{1}{n-i+1} T^{i-1} (1 - T^{n-i+1}) \\
&= \sum_{i=1}^n \frac{1}{n-i+1} T^{i-1} - \sum_{i=1}^n \frac{1}{n-i+1} T^n \\
&= T^n \sum_{i=1}^n \frac{1}{i} (T^{-i} - 1)
\end{aligned}$$

We take the limit $n \rightarrow \infty$, we have

$$\lim_{n \rightarrow \infty} T^n \sum_{i=1}^n \frac{1}{i} (T^{-i} - 1) = \frac{1}{e} \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{1}{i} \left(\left(1 + \frac{1}{n}\right)^i - 1 \right)$$

Observe that

$$\left(1 + \frac{1}{n}\right)^i - 1 = \sum_{k=1}^i \binom{i}{k} \frac{1}{n^k}.$$

Then, we take the sum on i , we have

$$\sum_{i=1}^n \frac{1}{i} \left(1 + \frac{1}{n}\right)^i - 1 = \sum_{i=1}^n \frac{1}{i} \sum_{k=1}^i \binom{i}{k} \frac{1}{n^k}$$

By Fubini Theorem, we have

$$\sum_{i=1}^n \frac{1}{i} \sum_{k=1}^i \binom{i}{k} \frac{1}{n^k} = \sum_{k=1}^n \sum_{i=k}^n \frac{1}{i} \binom{i}{k} \frac{1}{n^k}$$

We have

$$\frac{1}{i} \binom{i}{k} = \frac{(i-1)(i-2)\dots(i-k+1)}{k!}$$

The nominator is a polynomial with order $k-1$ with respect to i .

Exercise 2.1. Show that $\sum_{i=1}^n i^k = \frac{1}{k+1} n^{k+1} + O(n^k)$

By this property, we have

$$\lim_{n \rightarrow \infty} \sum_{i=k}^n \frac{1}{i} \binom{i}{k} \frac{1}{n^k} = \frac{1}{k \cdot k!} \frac{n^k}{n^k} = \frac{1}{k \cdot k!}.$$

Finally, we have the lower bound of $P^{\max}$ is

$$\frac{1}{e} \sum_{k=1}^n \frac{1}{k \cdot k!} \approx 0.485.$$

2.2 Upper Bound of $P^{\max}$

Consider there are only two secretaries in the arriving sequence. We have $X_1 \sim \text{Unif}(0, 1)$, $X_2 \sim \text{Unif}(0, 1)$.

Exercise 2.2. Show that the upper bound of $P^{\max}$ among all deterministic algorithms is $\frac{1}{2}$

Hint: Any deterministic algorithm can only choose X_1 or X_2 . If $X_1 = x$, what will happen? Then take integral. Another method is that think about a coupling random arrival instance $Y_1 = X_1$, $Y_2 = \text{Unif}(0, 1)$. Y obviously follows the arrival process distribution. When the first customer arrives, any algorithm cannot distinguish the arrival instance is X or Y . Then for the first customer, if the algorithm accepts him, ..., if not accept,....

Exercise 2.3. Show that the upper bound of $P^{\max}$ among all randomized algorithms is $\frac{1}{2}$

Hint: Yao's principle. (or set a probability distribution on X_1 and X_2).

Lecture 3: response to exercises

Mon 07 Feb 2022 00:24

2.3 Responses

2.3.1 Response to (2.1)

Proof. Use induction on n . The base case is trivial:

$$1 = \frac{1}{k+1} + O(1).$$

Now suppose the property holds for n .

$$\begin{aligned} \sum_{i=1}^{n+1} i^k &= (n+1)^k + \sum_{i=1}^n i^k = (n+1)^k + \frac{1}{k+1} n^{k+1} + O(n^k) \\ \frac{1}{k+1} (n+1)^{k+1} &= \frac{1}{k+1} n^{k+1} + O(n^k). \end{aligned}$$

By considering $(n+1)^k = O(n^k)$ and $O(n^k) \pm O(n^k) = O(n^k)$, the difference of the right hand side of the above equations is $O(n^k)$. The left hand side shows the property for $n+1$. $\square$

2.3.2 Response to (2.2)

Optimal P for a deterministic algorithm is 0.75.

Proof. ALG is completely characterized by a function $\alpha : [0, 1] \rightarrow [0, 1]$, where $\alpha(x)$ represents the probability of accepting X_1 after an observation $X_1 = x$.

$$\begin{aligned} P[\text{win}] &= \int_0^1 \alpha P[X_2 < x] + (1 - \alpha) P[X_2 > x] f(x) dx \\ &= \int_0^1 \alpha x + (1 - \alpha)(1 - x) \\ &= \int_0^1 2\alpha x - \alpha dx + \frac{1}{2} \\ &\leq \max_{\alpha} \int_0^1 2\alpha x - \alpha dx + \frac{1}{2}. \end{aligned}$$

ALG is deterministic iff $\alpha(x) \in \{0, 1\}$. If α takes 0, the integral is zero. If α takes 1, $2\alpha x - \alpha > 0$ iff $x > \frac{1}{2}$. Hence to maximize the integral we take

$$\alpha(x) = \begin{cases} 1 & x \geq \frac{1}{2} \\ 0 & x < \frac{1}{2} \end{cases}, \text{ which gives an upper bound of } \frac{3}{4}. \quad \square$$

Response to (2.3)

Lecture 4: Relaxed SP with error

Tue 01 Feb 2022 00:06

3 Calculation with error involved

The weights $X_1, \dots, X_n \sim \text{Unif}(0, 1)$. The noisy observations $\tilde{X}_1, \dots, \tilde{X}_n \sim \text{Unif}(x_i - \delta, x_i + \delta)$.

The strategy is to introduce thresholding parameters R and $T \in (0, 1)$. We examine the upcoming item i if $\tilde{x}_i > R$, and accept the item if $x_i > T$. We still want to find $\mathbb{P}[X_{max} \text{ is selected}]$ given particular R, T, n, δ .

$$\mathbb{P}[X_{max} \text{ selected}] = \sum_i \int_0^1 \mathbb{P}[X_i \text{ best} \cap i \text{ selected} \mid X_i = x] f(x) dx \quad (15)$$

$$\begin{aligned} & \mathbb{P}[X_i \text{ best} \cap i \text{ selected} \mid X_i = x] \\ &= \mathbb{P}[\text{reach } i \cap \tilde{X}_i > R \cap X_i > T \cap X_i \text{ max} \mid X_i = x] \\ &= \mathbb{P}[\text{Unif}(x - \delta, x + \delta) > R] \mathbf{1}(x > T) \mathbb{P}[\text{reach } i \cap X_i \text{ max} \mid X_i = x] \end{aligned}$$

$$\begin{aligned} & \mathbb{P}[\text{reach } i \cap X_i \text{ max} \mid X_i = x] \\ &= \sum_{k=0}^{\lfloor \frac{i-1}{2} \rfloor} \mathbb{P}[\text{reach } i \cap X_i \text{ max} \cap k \text{ examines before reaching } i \mid X_i = x] \\ &= \sum_{k=0}^{\lfloor \frac{i-1}{2} \rfloor} \mathbb{P}[\text{reach } i \text{ with } k \text{ examines}] \mathbb{P}[X_i \text{ max} \mid \text{reach } i \text{ with } k \text{ examines} \cap X_i = x] \\ &= \sum_{k=0}^{\lfloor \frac{i-1}{2} \rfloor} \underbrace{\frac{(i-1-k)!}{k!(i-1-2k)!}}_{\text{ways to reach } i} \mathbb{P}[\text{ED}]^k \mathbb{P}[\text{D}]^{i-1-2k} \underbrace{F(x \mid X < T)^k}_{\text{examined}} \underbrace{F(x)^k}_{\text{omitted}} \underbrace{F(x \mid \tilde{X} < R)^{i-1-2k}}_{\text{declined directly}} \underbrace{F(x)^{n-i}}_{\text{upcoming}} \end{aligned}$$

where

$$\begin{aligned} \mathbb{P}[\text{ED}] &= \frac{1}{2\delta} \int_0^T \int_{y-\delta}^{y+\delta} \mathbf{1}(z > R) dz dy = \text{const} \\ \mathbb{P}[\text{D}] &= \frac{1}{2\delta} \int_0^1 \int_{y-\delta}^{y+\delta} \mathbf{1}(z < R) dz dy = \text{const} \end{aligned}$$

and for $0 \leq x \leq 1$:

$$\begin{aligned} F(x \mid X < T) &= \begin{cases} \frac{x}{T} & 0 \leq x < T \\ 1 & x \geq T \end{cases} \\ F(x \mid \tilde{X} < R) &= \frac{\int_0^x \int_{y-\delta}^{y+\delta} \mathbf{1}(z < R) dz dy}{\int_0^1 \int_{y-\delta}^{y+\delta} \mathbf{1}(z < R) dz dy} \\ F(x) &= x. \end{aligned}$$